Milestone M2 Package P06 (06_identifiability): Geometry & Station Confounding Report

Project: IIT/SRIC/R/AEH/2026/104 (CMPDI) — *Dynamic Modelling and Digital Twin of Electric Cable Shovel* **Package ID:** MS02P06 (M2/experiments/06_identifiability) **Primary Verified Archives:** M2_100kN_0p1111Hz_s-1p32.mat, M2_wrongstation_narrowband.mat, M2_wrongstation_bi_band.mat, and M2_sensitivity_diagnostic.mat.

1. Reconciliation of All Handle Station Definitions (r_{COG}, s_{bail}, s_{tip})

Across M1, M2, and M3, all axial stations along the dipper handle are measured radially outward from the **saddle pivot axis (O_3)** as a function of the prismatic crowd extension $d_4(t)$ (which measures the distance from the saddle pivot O_3 to the handle reference datum $x_{\mathrm{ref}} = 1.50\ \mathrm{m}$ upstream of the bail pin):

Station Symbol	Formula in Terms of $d_4(t)$	Physical Location on P&H 2100BL Handle--Dipper Assembly	Range over Wide Stroke $d_4 \in [6.7505, 10.7495]\ \mathrm{m}$	Range over BI Band $d_4 \in [9.5001, 10.0399]\ \mathrm{m}$	Where Used in Code & Models
$r_{\mathrm{COG}}(d_4)$	$d_4 + c = d_4 - 1.32\ \mathrm{m}$ ($c = x_{\mathrm{ref}} - L_{\mathrm{COG}} = 1.50 - 2.82 = -1.32\ \mathrm{m}$)	Composite center of mass (COG) of the dipper + handle assembly (Rasuli et al., 2014, Table II)	$[5.4305,\ 9.4295]\ \mathrm{m}$	$[8.1801,\ 8.7199]\ \mathrm{m}$	D(1,1) = I_zz + M_d*r^2, C, G(1) = M_d*g*r*cos(q3) in shovel_params.m, recover_load.m, and m3_observer_suite.m
$s_{\mathrm{bail}}(d_4)$	$d_4 + x_{\mathrm{ref}} = d_4 + 1.50\ \mathrm{m}$	Hoist rope attachment (bail pin) on the handle axis (eccentricity e = 0, Assumption A10 in M2_README.	$[8.2505,\ 12.2495]\ \mathrm{m}$	$[11.0001,\ 11.5399]\ \mathrm{m}$	External force injection frame in M2_LoadInjection.slx, Station = +1.50 in recover_load.m, and task-space

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		md)			Jacobian J_s(q) in m3_observer_suite.m
s_{\mathrm{wrong}}(d_4)	d_4 - 1.32\ \mathrm{m} \equiv r_{\mathrm{COG}}(d_4)	Deliberate wrong-station negative control (Station = -1.32) testing whether a bail load can be mistaken for a COG load or dipper mass error	[5.4305,\ 9.4295]\ \mathrm{m}	[8.1801,\ 8.7199]\ \mathrm{m}	Used in M2_100kN_0p1111Hz_s-1p32.mat, M2_wrongstation_narrowband.mat, and M2_wrongstation_bi_band.mat
s_{\mathrm{tip}}(d_4)	d_4 + x_{\mathrm{tip}} (future M7 extension)	Dipper tooth-tip cutting edge (used in Feng et al., 2022 and Raza & Frimpong, 2017)	Deferred to M7	Deferred to M7	Documented in Revision V3 & V23 of M2_design_evolution.md

Audit Note on Superseded M3 v1.0 Offset: In the superseded v1.0 draft of m3_observer_suite.m, a synthetic station r_bail = d4 + 2.05 m over d_4 \in [2.6, 4.1]\ \mathrm{m} had been hard-coded. That discrepancy (AUD-06) was completely eliminated in v2.0.0-PRELIMINARY, which locks s_bail = d4 + 1.50 m and r_COG = d4 - 1.32 m identically to shovel_params.m and recover_load.m.

2. Analytical Derivation of Station Confounding (\rho_{\mathrm{station}}(d_4))

When an external vertical force F_y(t) is injected into the Simscape plant at the bail station s_{\mathrm{bail}}(t) = d_4(t) + 1.50\ \mathrm{m}, the true saddle residual is:
e_{\tau_3}(t) = \tau_{3,\mathrm{meas}}(t) - \tau_{3,\mathrm{model}}(t) = -(d_4(t) + 1.50) F_y(t)\cos q_3(t).
If an analyst or estimator mistakenly assumes the force acts at the COG station r_{\mathrm{COG}}(t) = d_4(t) - 1.32\ \mathrm{m} (or equivalently attempts to explain the residual via a quasi-static dipper mass error \Delta M_d g(d_4(t) - 1.32)\cos q_3(t)), the predicted reference torque is:
Q_{q_3,\mathrm{wrong}}(t) = (d_4(t) - 1.32) F_y(t)\cos q_3(t).
The pointwise ratio of the true residual magnitude to the wrong-station prediction is strictly

independent of q_3(t) and F_y(t):
\rho_{\mathrm{station}}(d_4) = \frac{\vert{}e_{\tau_3}(t)\vert{}}{\vert{}Q_{q_3,\mathrm{wrong}}(t)\vert{}} = \frac{d_4 + 1.50}{d_4 - 1.32} = 1 + \frac{2.82}{d_4 - 1.32}, \qquad \frac{\mathrm{d}\rho_{\mathrm{station}}}{\mathrm{d}d_4} = -\frac{2.82}{(d_4 - 1.32)^2}.
Consequently:

1. Over any trajectory where d_4(t) \approx d_{4,\mathrm{mid}} varies over a narrow span \Delta d_4, e_{\tau_3}(t) is almost perfectly proportional to Q_{q_3,\mathrm{wrong}}(t) with regression slope -\rho_{\mathrm{station}}(d_{4,\mathrm{mid}})!
2. A linear regression with free slope and intercept therefore achieves R^2 \to 1, **masking the 2.82\ \mathrm{m} station error behind an inflated force/mass scale factor** (\approx 1.335\times to 1.381\times).
3. Meanwhile, on the prismatic **crowd channel (d_4)**, Q_{d_4}(t) = F_y(t)\sin q_3(t) is completely independent of the handle station s, so NRMSE_d4 remains at 5.07\times 10^{-14}\% even when the station is wrong by 2.82\ \mathrm{m}!

3. Verified Quantitative Results Across All Three Wrong-Station .mat Archives

Wrong-Station Experiment (Station = -1.32 m)	Archive & SHA-256 Prefix	Realized d_4 Range (\mathrm{m})	\Delta d_4 Span (\mathrm{m})	Analytical \rho(d_{4,\mathrm{mid}})	Measured Regression \vert{}\text{Slope}\vert{} (Diff \%)	Saddle R^2	Discriminating Fraction 1 - R^2	Unscaled Saddle \mathrm{NRMSE}_{q_3} (\%)
Wide Stroke (d4 = 6.75..10.75 m, span 4.00 m)	M2_100kN_0p1111Hz_s-1p32.mat (3d216bba7f88b2af)	[6.7505,\ 10.7495]	3.9990	1.3795	1.3785 (0.079\%)	0.99730824	2.692e-03	13.72\%
Centered Narrowband (d4 = 8.48..9.02 m, span 0.54 m)	M2_wrongstation_narrowband.mat (6fe32dbbba80f895)	[8.4801,\ 9.0199]	0.5399	1.3795	1.3809 (0.102\%)	0.99995305	4.695e-05	11.71\%
Bi et al. Operational Band (d4 = 9.50..10.04 m, span 0.54 m)	M2_wrongstation_bi_band.mat (5e2a85343715d45e)	[9.5001,\ 10.0399]	0.5399	1.3337	1.3348 (0.083\%)	0.99997001	2.999e-05	10.27\%

Key Takeaway (Headline Result for M2 Figure 1 & Figure 4 in plot_m2.m)

- Restricting the crowd stroke from the wide identification envelope (\Delta d_4 = 3.9990\ \mathrm{m}) to the Bi et al. (2020) digging band (\Delta d_4 = 0.5399\ \mathrm{m}) collapses the non-collinear discriminating fraction 1 - R^2 from 2.692\times 10^{-3} to 2.999\times 10^{-5} — a **89.8\times collapse** (and a **57.3\times collapse** on the centered narrowband d_4 = 8.75 \pm 0.27\ \mathrm{m} where 1 - R^2 = 4.695\times 10^{-5}).
- Across all three runs, the measured regression slope matches the analytical curve \rho(d_{4,\mathrm{mid}}) = (d_{4,\mathrm{mid}} + 1.50)/(d_{4,\mathrm{mid}} - 1.32) to within **0.072\%\text{--}0.104\%**.